METRIC

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PERFORMANCE SPECIFICATION

# GREASE, AIRCRAFT AND INSTRUMENT, GEAR AND ACTUATOR SCREW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for one grade of grease for use at extremely low temperatures on aircraft and instruments. This grease is effective in the temperature range from -73° to 121°C. The type I grease is identified by NATO code Number G-354 (see 6.8).

1.2 Classification. The grease will be of the following types:

Type I – Grease thickened with a metallic soap. NATO Code Number G-354.
Type II - Grease thickened with clay.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 491000B120-3, Highway 547, Lakehurst, NJ 08733-5100 or emailed to 4Hthomas.omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at 5Hhttp://assist.daps.dla.mil



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2.2 Government documents.

2.2.1 Standard. The following standard forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of this document are those cited in the solicitation or contract.

FEDERAL STANDARDS

| | | |
|---|---|---|
| FED-STD-791 | - | Lubricants, Liquid Fuels, and Related Products; Methods of Testing. |

(Copies of this document is available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) INTERNATIONAL

| | | |
|---|---|---|
| ASTM-D217 | - | Cone Penetration of Lubricating Grease, Standard Test Methods for (DoD adopted) |
| ASTM-D942 | - | Oxidation Stability of Lubricating Greases by the Oxygen Bomb Method, Standard Test Method for (DoD adopted) |
| ASTM-D1264 | - | Determining the Water Washout Characteristics of Lubricating Greases, Standard Test Method for (DoD adopted) |
| ASTM-D1478 | - | Low-Temperature Torque of Ball Bearing Grease, Standard Test Method for (DoD adopted) |
| ASTM-D1743 | - | Determining Corrosion Preventative Properties of Lubricating Greases, Standard Test Method for (DoD adopted) |
| ASTM-D2265 | - | Dropping Point of Lubricating Grease Over Wide Temperature Range Standard Test Method for (DoD adopted) |
| ASTM-D2595 | - | Evaporation Loss of Lubricating Greases Over Wide Temperature Range, Standard Test Method for (DoD adopted) |



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| | | |
|---|---|---|
| ASTM-D2596 | - | Measurement of Extreme Pressure Properties of Lubricating Grease (Four Ball Method), Standard Test Method for (DoD adopted) |
| ASTM-D3336 | - | Life of Lubricating Greases in Ball Bearings at Elevated Temperatures, Standard Test Method for (DoD adopted) |
| ASTM-D4048 | - | Detection of Copper Corrosion from Lubricating Grease, Standard Test Method for (DoD adopted) |
| ASTM-D4057 | - | Manual Sampling of Petroleum and Petroleum Products, Standard Practice for (DoD adopted) |
| ASTM-D6184 | - | Oil Separation from Lubricating Grease (Conical Sieve Method), Standard Test Method for (DoD Adopted) |

(Copies of these documents are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Qualification. The grease furnished under this specification shall be a product that is authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3). Any change in the formulation of a qualified product shall require requalification of that product.

3.2 Material. The grease shall consist of a gelling agent, a diester base oil, and an extreme-pressure additive, and any other chemical addition agent required to meet the requirements of this specification (see 6.7) Type I and type II greases shall not be mixed together.

3.2.1 Toxicity. The grease furnished under this specification shall have no adverse effect on the health of personnel when used for its intended purpose (see 6.2).

3.3 Physical properties. Physical properties of the grease shall be in accordance with table I.



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TABLE I. Physical properties.

| Property | Requirements |
|---|---|
| Corrosiveness (copper strip), maximum 1/ | 1b |
| Dirt, particles per ml of grease, maximum: | |
| 25 – 74 micrometres diameter | 1,000 |
| 75 micrometres diameter or larger | none |
| Dropping point, °C, minimum | 165 |
| Evaporation, percent, weight loss in 22 hours at 100° ±0.5°C, maximum | 2 |
| Gear wear, mg/1,000 cycles, maximum: | |
| under a 2.3 kg load | 2.5 |
| under a 4.5 kg load | 3.5 |
| High temperature performance, hours, at 121°C, minimum | 2/ 1,000 |
| Load carrying capacity, mean Hertz load, minimum | 30 |
| Low temperature torque at -73° ±1°C: | |
| starting, Nm, maximum | 1.00 |
| running (after 60 minutes), Nm, maximum | 0.10 |
| Odor | No odor of rancidity, perfume, or free alcohol |
| Oil separation, percent, weight loss in 30 hours, maximum | 5 |
| Oxidation stability: | |
| Bomb oxidation, pressure drop, Kpa, maximum: | |
| in 100 hours | 70.00 |
| in 500 hours | 105.0 |
| Penetration, (0.1 mm): | |
| unworked, minimum | 200 |
| worked | 270 – 310 |



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TABLE I. Physical properties – Continued.

| Property | Requirement |
|---|---|
| Rust preventative properties, maximum passing rate | 2 |
| Storage stability at 40° ±2°C , penetration (0.1 mm) after: | |
| unworked, minimum | 200 |
| worked (change from original), maximum | 30 |
| Water resistance, at 38° ±3°C, percent lubricant washed out, maximum | 20 |
| Worked stability, penetration (0.1 mm) | 270 – 375 |

1/ The grease shall show no green color in that portion contacting the copper strip.
2/ Average of four runs.

3.4 Workmanship. The grease, when examined visually, shall be a smooth homogenous mixture, free of lumps and abrasive material. When worked with a spatula on a glass surface, the grease shall exhibit uniformity and ability to spread with a straight edge to a smooth surface.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the conditions specified in tables I and II on unworked grease. The physical requirements specified in table I apply to the average of determinations made on the sample.

4.3 Qualification inspection. Qualification inspection shall consist of all the requirements and tests in table II.

4.3.1 Sampling. Qualification test samples shall consist of a minimum of 5 kilograms (kg) of grease for which qualification is desired.


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TABLE II. Qualification inspection and methods.

| Requirements | Test method or paragraph |
|---|---|
| Corrosiveness (copper strip) | 1/ ASTM-D4048 |
| Dirt | FED-STD-791, Method 3005 |
| Dropping point | ASTM-D2265 |
| Evaporation | ASTM-D2595 |
| Gear wear | FED-STD-791, Method335 |
| High temperature performance | ASTM-D3336 |
| Load carrying capacity | ASTM-D2596 |
| Low temperature torque 2/ | ASTM-D1478 |
| Odor | 4.5.1 |
| Oil separation | ASTM-D6184 |
| Oxidation stability | ASTM-D942 |
| Penetration | ASTM-D217 |
| Rust preventative properties | ASTM-D1743 |
| Storage stability 3/ | FED-STD-791, Method 3467 |
| Water resistance | ASTM-D1264 |
| Worked stability | FED-STD-791, Method 313 |
| Workmanship | Visual |

1/ The copper strip shall not tarnish more than a classification of 1b when compared with ASTM copper strip corrosion standards. Test shall be conducted at 100° ±1°C for 24-hours.
2/ Test shall be conducted at -73° ±1°C. Observation for skidding or ball sliding shall be made during test run.
3/ Temperature of 40° ±2°C to be maintained for 6 months.


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4.4 Conformance inspection.

4.4.1 Sampling. Conformance inspection sample shall consist of two, 2 kg samples of grease taken at random from filled containers from each lot of grease. For users who obtain grease in large containers, samples shall be selected in accordance with ASTM-D4057. The sample shall be tested as specified in 4.4.2.

4.4.2 Conformance testing. Conformance samples selected in accordance with 4.4.1 shall be subjected to the following tests performed in accordance with the applicable test methods in table II, and shall meet the applicable requirements of table I.

| | |
|---|---|
| Corrosiveness (copper strip) | Odor |
| Dirt | Oil separation |
| Dropping point | Penetration |
| Evaporation | Rust preventative properties |
| Load carrying capacity | Water resistance |
| Low temperature torque | Worked stability |

Failure of either conformance sample to comply with any of the tests listed above shall be cause to reject the lot represented by these samples.

4.5 Test methods. Test methods shall be as specified in table II.

4.5.1 Odor. The odor of the grease shall be observed for conformance to table I immediately upon opening the test containers.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory).



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6.1 Intended use. The grease is intended for use in ball, roller and needle bearings, gears, and on sliding and rolling surfaces of such equipment as instruments, cameras, electronic gear and aircraft control systems that are subject to extreme marine and low temperature conditions not encountered in commercial aviation. It can be used for rolling and sliding surfaces of equipment having low motivating power (low torque equipment). Its extremely low volatility (see evaporation requirement in table I) is of advantage in preventing oil fogging in aircraft optical instruments. This grease is also intended for use on aircraft gears, actuator screws, and other equipment requiring a lubricant with high load carrying capacity over a temperature range of -73° to +121°C.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity and type.

c. Toxicity (see 6.5).

d. Packaging requirements and marking (see 5.1 and 6.2.1).

6.2.1 Marking. Containers should be marked in accordance with MIL-STD-290, "Packaging of Petroleum and Related Products" and with the cautionary statement DO NOT MIX TYPE I AND TYPE II MATERIAL TOGETHER. All unit and intermediate packs of toxic and hazardous chemicals and materials should also be labeled in accordance with the applicable laws, statutes, regulations, and ordinances, including federal, state, and municipal requirements. Unit and intermediate containers should be marked in accordance with ANSI-Z129.1, "Precautionary Labeling of Hazardous Industrial Chemicals."

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-23827, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Commander, Attn: Code 4.3.4.1, Building 2188, Naval Air Warfare Center Aircraft Division, 22347 Cedar Point Road, Unit 6, Patuxent River, MD 20670-1161.

6.3.1 Submission of samples. Prior to submitting samples for qualification testing, manufacturers will request authorization from the qualification activity. Upon receipt of authorization, the samples will be forwarded as directed. The qualifying activity will require the manufacturer to submit for review and approval, two copies of the manufacturer's test report, including the sample's Material Safety Data Sheet (MSDS) (see 6.5), the location and identity of


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the plant which produced the sample, and the item composition report. The samples should be plainly and durably marked with the following information:

Sample for qualification inspection.
GREASE, AIRCRAFT AND INSTRUMENT, GEAR AND ACTUATOR SCREW
Name of manufacturer.
Product code number.
Batch number.
Date of manufacture.
Submitted by (name) (date) for qualification inspection in accordance with MIL-PRF-23827C under authorization of (reference authorizing letter) (see 6.3).

6.3.2 Retention of qualification. In order to retain qualification of a product approved for listing on the Qualified Products List (QPL), the Government will require the manufacturer to verify by certification to the qualifying activity that the manufacturer's product complies with the requirements of this specification. The time of periodic verification by certification will be in two year intervals from the date of original qualification and will be initiated by the Government. The Government reserves the right to re-examine the qualified product whenever deemed necessary to ensure that the product continues to meet any or all of the specification requirements.

6.4 Lot and batch. A lot is the grease produced by one manufacturer at one plant from the same materials and under essentially the same conditions, provided the operation is continuous and does not exceed a 24-hour period. In the event the process is a batch operation, each batch constitutes a lot. A batch is that quantity of material which has been manufactured by the same unit chemical process and subjected to the same physical mixing operation intended to make the final product substantially uniform.

6.5 Material safety data sheets (MSDSs). Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313; and 29 CFR 1910.1200 requires that the Material Safety Data Sheet for each hazardous chemical used in an operation must be readily available to personnel using the material.

6.6 Subject term (key word) listing.

High load carrying
Low temperature
Lubricant


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6.7 Composition for volatility and low temperature. The composition if the lubricant is not otherwise limited, but substantial proportions of non-petroleum materials will be required to meet the volatility and low temperature requirements of this specification (see 3.2).

6.8 International standardization agreements. International standardization agreement implementation. This specification implements ASCC AIRSTD 15/1, "Interchangeability Chart of Standardized Aviation Fuels, Lubricants and Allied Products" and NATO STANAG 1135, "Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations.". When amendment, revision, or cancellation of this specification is proposed the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at http://assist.daps.dla.mil.

6.9 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

## CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - AS
Air Force – 11
DLA - GS

Preparing activity:
Navy - AS
(Project 9150-2006-005)

Review activities:
Army - AV, EA, MI
Navy - OS
Air Force - 68

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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